

Part A: Fill in the Blanks

- ✓ 1. A router forwards packets between different _____. Answer: Networks
- ✓ 2. A router mainly operates at the _____ layer of the OSI model. Answer: Network
- ✓ 3. A router selects the _____ path to a destination network. Answer: Best
- ✓ 4. The running configuration of a router is stored in _____. Answer: RAM
- ✓ 5. The startup configuration is stored in _____. Answer: NVRAM
- ✓ 6. Cisco IOS is generally stored in _____ memory. Answer: Flash
- ✓ 7. POST and bootstrap code are stored in _____ memory. Answer: ROM
- ✓ 8. A host uses a _____ to communicate with devices outside its local network. Answer: Default Gateway
9. An IPv4 address consists of _____ bits. Answer: 32 i won
10. An IPv6 address consists of _____ bits. Answer: 128
11. The command used to activate a router interface is _____. Answer: no shutdown
12. The command used to enter Privileged EXEC Mode is _____. Answer: enable
13. The command used to enter Global Configuration Mode is _____. Answer: configure
terminal
14. The command used to change the router name is _____. Answer: hostname
15. The command used to configure the console line is _____. Answer: line console 0
16. The command used to configure a privileged EXEC password is _____. Answer: enable secret
17. The command used to configure an IPv4 address is _____. Answer: ip address
18. The command used to configure an IPv6 address is _____. Answer: ipv6 address
19. IPv6 routing is enabled using the command _____. Answer: ipv6 unicast-routing

20. The command used to display brief interface information is _____. Answer: show ip interface brief:-

21. The command used to display the IPv4 routing table is _____. Answer: show ip route

22. The command used to display the IPv6 routing table is _____. Answer: show ipv6 route

23. A route used when no specific route is available is called a _____ route. Answer:

Default

24. A route manually configured by an administrator is called a _____ route. Answer: Static

25. A route learned through a routing protocol is called a _____ route. Answer: Dynamic

26. The routing table code for a directly connected route is _____. Answer: C

27. The routing table code for a local route is _____. Answer: L

28. The routing table code for a static route is _____. Answer: S

29. The routing table code for RIP is _____. Answer: R

30. The routing table code for OSPF is _____. Answer: O

31. The routing table code for EIGRP is _____. Answer: D

32. A routing protocol uses a _____ to select the best route. Answer: Metric

33. The reliability of different routing sources is determined by _____. Answer:

Administrative Distance

34. Using multiple routes with the same metric is called _____. Answer: Equal-Cost Load

Balancing

35. Removing the old Layer 2 header from a frame is called _____. Answer: De-encapsulation

36. Adding a new Layer 2 header to a packet is called _____. Answer: Re-encapsulation

37. _____ is used to find a MAC address from an IPv4 address. Answer: ARP

Loopback

38. A _____ interface is a logical interface used for testing and management. Answer:

Loopback

39. SVI stands for _____. Answer: Switched Virtual Interface

40. The process by which routers achieve a consistent routing view after a network change is called _____. Answer: Convergence

41. Selecting the most specific route is called _____. Answer: Longest Prefix Match

42. The current active configuration is called the _____ configuration. Answer: Running

43. The configuration loaded after a router restarts is called the _____ configuration.

Answer: Startup

Startup

44. The command used to save the running configuration is _____. Answer: copy

running-config startup-config

45. The interface through which a router sends a packet is called the _____ interface.

Answer: Exit

Exit

46. The address of the next router in the path is called the _____ address. Answer: Next-

Hop

47. A routing table is a database containing information about _____. Answer: Routes

48. If a router has no matching route or default route, it may _____ the packet. Answer:

Drop

49. The metric used by RIP is _____. Answer: Hop Count

50. The metric used by OSPF is _____. Answer: Cost

Part B: Short Questions

1. What is a router?

A router is a Layer 3 networking device that connects different networks and forwards packets toward their destination.

2. Write down the main functions of a router.

1. Connecting different networks
2. Maintaining a routing table
3. Selecting the best path
4. Forwarding packets
5. Controlling communication between networks

3. Differentiate between a router and a switch.

Router	Switch
Connects different networks	Connects devices within a LAN
Works mainly at Layer 3	Works mainly at Layer 2
Uses IP addresses	Uses MAC addresses
Uses a routing table	Uses a MAC address table

4. What is the function of RAM in a router?

RAM stores the running configuration, routing table, ARP cache, and temporary packet information.

5. What is the function of ROM?

ROM stores POST, bootstrap code, and basic startup instructions.

6. What is the function of NVRAM?

NVRAM stores the startup configuration, which is loaded when the router restarts.

7. What is Flash Memory?

Flash Memory stores the Cisco IOS software used by the router.

8. What is a default gateway?

A default gateway is the router interface address used by a host to communicate with devices outside its local network.

9. What is DHCP?

DHCP is a protocol that automatically provides hosts with IP addresses, subnet masks, default gateways, and other network information.

10. What is a static IP address?

A static IP address is an IP address manually assigned to a device by an administrator.

11. What is console access?

Console access is the process of configuring a router locally through its console port using a console cable.

12. What is an SVI?

An SVI, or Switched Virtual Interface, is a logical interface on a switch, commonly used for management purposes.

13. What is a loopback interface?

A loopback interface is a logical interface that does not depend on a physical port. It is commonly used for testing, management, and router identification.

14. What is the function of the `no shutdown` command?

The `no shutdown` command activates a router interface.

15. What is the function of `show ip interface brief`?

It displays brief information about router interfaces, including IP addresses and interface status.

16. What is a routing table?

A routing table is a database that contains information about destination networks and the paths used to reach them.

17. What is a routing decision?

A routing decision is the process of examining a destination IP address and selecting the appropriate route from the routing table.

18. What is a next-hop address? → next router er ip address

A next-hop address is the IP address of the next router to which a packet is forwarded.

19. What is an exit interface? ei interface router er packet leave korar jonno de

An exit interface is the router interface through which a packet leaves the router.

20. What is a metric? available route er modde best route

A metric is a value used by a routing protocol to select the best route among multiple available routes.

21. What is administrative distance? reliability of routers value

Administrative distance is a value used to determine the reliability of routes learned from different routing sources. A lower value is generally preferred.

22. What is static routing? manually configure

Static routing is a routing method in which routes are manually configured by a network administrator.

23. What is dynamic routing? automatically learn and update

Dynamic routing is a routing method in which routers automatically learn and update routes using routing protocols.

24. Give three examples of routing protocols.

1. RIP
2. OSPF
3. EIGRP

25. What is convergence?

Convergence is the process by which all routers learn and agree on the updated network topology after a network change.

26. What is equal-cost load balancing?

Equal-cost load balancing is the process of forwarding traffic through multiple routes that have the same metric to the same destination.

27. What is packet forwarding?

Packet forwarding is the process of sending a packet through the appropriate next hop or exit interface based on the routing table.

28. What is de-encapsulation?

De-encapsulation is the process of removing the old Data Link Layer header and trailer from a received frame.

29. What is re-encapsulation?

Re-encapsulation is the process of adding a new Data Link Layer header and trailer before forwarding a packet through another network.

30. What is the function of ARP?

ARP finds the MAC address associated with an IPv4 address.

31. What is a directly connected route?

A directly connected route is automatically created when an IP address is configured on an active router interface.

32. What is a local route?

A local route represents the specific IP address configured on a router interface.

33. What is a default route?

A default route is used when no more specific route exists for a destination network.

34. Write two differences between IPv4 and IPv6.

IPv4	IPv6
<u>32-bit address</u>	<u>128-bit address</u>
Uses <u>decimal notation</u>	Uses <u>hexadecimal notation</u>
Uses <u>ARP</u>	Uses <u>Neighbor Discovery</u>
<u>Smaller address space</u>	<u>Much larger address space</u>

35. Differentiate between running configuration and startup configuration.

- Running Configuration: The active configuration stored in RAM.
- Startup Configuration: The saved configuration stored in NVRAM.

36. What are the two parts of interface status?

1. Line status
2. Protocol status



37. Why is the show ip route command important?

It displays the known networks, route sources, next-hop addresses, and exit interfaces in the router's routing table.

38. Why is a default route important?

A default route allows a router to forward packets when there is no specific route for the destination network.

39. What is longest prefix match?

Longest prefix match is the process of selecting the most specific matching route from the routing table.

40. Which addresses change during packet forwarding?

The Layer 2 source and destination addresses change at each network link. The source and destination IP addresses generally remain unchanged.

Part C: Broad Questions

1. Discuss the functions and characteristics of a router.

Answer:

A router is a Layer 3 networking device that connects different networks and forwards packets from one network to another. It examines the destination IP address of a packet and selects the best path using its routing table.

Main functions of a router:

1. Connecting different networks
2. Maintaining routing information
3. Selecting the best path
4. Forwarding packets
5. Separating broadcast domains
6. Controlling communication between networks
7. Supporting different routing protocols

Important characteristics:

- Speed
- Cost
- Security
- Availability
- Scalability
- Reliability

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Therefore, routers play an important role in providing efficient and controlled communication between networks.

2. Explain the functions of RAM, ROM, NVRAM, and Flash Memory.

Answer:

Memory	Function
RAM	Stores <u>running configuration</u> , <u>routing table</u> , <u>ARP cache</u> , and <u>temporary data</u>
ROM	Stores <u>POST</u> , <u>bootstrap code</u> , and <u>startup instructions</u>
NVRAM	Stores <u>startup configuration</u>
Flash	Stores <u>Cisco IOS software</u>

Explanation:

RAM: It stores temporary and active information. Its contents are lost when the router is powered off.

ROM: It contains the basic instructions required to start the router, including POST and bootstrap code.

NVRAM: It stores the startup configuration permanently so that it can be loaded after a restart.

Flash Memory: It stores the Cisco IOS software used to operate the router.

3. Describe the basic initial configuration of a router:-

Answer: :-

Step 1: Enter Privileged EXEC Mode

```
Router> enable
```

Step 2: Enter Global Configuration Mode

```
Router# configure terminal
```

Step 3: Set the Router Hostname

```
Router(config)# hostname R1
```

Step 4: Configure the Enable Secret Password

```
R1(config)# enable secret class
```

Step 5: Configure the Console Line

```
R1(config)# line console 0  
R1(config-line)# password cisco  
R1(config-line)# login  
R1(config-line)# exit
```

Step 6: Configure an Interface

```
R1(config)# interface gigabitEthernet 0/0  
R1(config-if)# ip address 192.168.1.1 255.255.255.0  
R1(config-if)# no shutdown  
R1(config-if)# exit
```

Step 7: Save the Configuration

```
R1# copy running-config startup-config
```

These steps configure the router's name, password, interface address, and interface status.

4. Explain the concept of a default gateway with an example.

Answer:

A default gateway is the router interface address used by a host to communicate with devices outside its local network.

Suppose a PC has the following configuration:

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

If the PC sends a packet to 192.168.1.20, the destination is in the same network, so the PC can communicate directly.

However, if the PC sends a packet to 192.168.2.10, the destination is in another network. Therefore, the PC forwards the packet to its default gateway, 192.168.1.1. The router then forwards the packet toward the destination network.

Importance:

1. Provides access to remote networks
2. Defines the exit point of a local network
3. Enables communication between different networks

5. Write the IPv4 and IPv6 interface configuration commands.

IPv4 Configuration:

```
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown
```

IPv6 Configuration:

```
Router(config)# ipv6 unicast-routing
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ipv6 address 2001:db8:1::1/64
Router(config-if)# no shutdown
```

Verification Commands:

```
show ip interface brief
show ipv6 interface brief
```

```
show interfaces
show ip route
show ipv6 route
```

The `ip address` command configures IPv4, while the `ipv6 address` command configures IPv6. The `no shutdown` command activates the interface.

6. Explain the routing table and routing decision process.

Answer:

A routing table is a database maintained by a router. It contains information about destination networks and the paths used to reach those networks.

Information stored in a routing table:

1. Route source
2. Destination network
3. Subnet mask or prefix
4. Administrative distance
5. Metric
6. Next-hop address
7. Exit interface

Routing decision process:

1. The router receives a packet.
2. It examines the destination IP address.
3. It searches the routing table for a matching route.
4. It selects the most specific route.
5. It determines the next-hop address or exit interface.
6. It forwards the packet.
7. It adds a suitable Layer 2 header before sending the packet.

If no matching route or default route exists, the router may drop the packet.

7. Compare static routing and dynamic routing.

Answer:

Static Routing: Routes are manually configured by a network administrator.

Example:

```
Router(config)# ip route 192.168.2.0 255.255.255.0 10.0.0.2
```

Dynamic Routing: Routes are automatically learned and updated through routing protocols.

Examples:

- RIP
- OSPF
- EIGRP

Differences:

Static Routing	Dynamic Routing
Manually configured	Learned through routing protocols
Manual updates are required	Routes can update automatically
Suitable for small networks	Suitable for large networks
Uses fewer resources	Uses more processing and bandwidth
Simple to configure	More complex to configure
Poor adaptability to failures	Better adaptability to topology changes

8. Explain metric and administrative distance.

Answer:

Metric:

A metric is used by a routing protocol to select the best route among multiple routes learned through the same protocol.

Examples:

- RIP uses hop count.
- OSPF uses cost.
- EIGRP uses a composite metric.

Administrative Distance:

Administrative Distance is used to determine the reliability of routes learned from different routing sources. A lower administrative distance is preferred.

Differences:

Metric	Administrative Distance
Compares routes within a routing protocol	Compares routes from different routing sources
Selects the best path	Selects the more reliable route source

Protocol-specific

Lower metric is generally preferred

Source-specific

Lower AD is generally preferred

9. Explain the packet forwarding process step by step.

Answer:

1. The router receives a frame through an incoming interface.
2. It removes the Data Link Layer header and trailer.
3. It examines the destination IP address.
4. It searches the routing table for the best route.
5. It determines the next-hop address or exit interface.
6. It may use ARP to find the next-hop MAC address.
7. It adds a new Data Link Layer header and trailer.
8. It forwards the packet to the next router or destination host.

The Layer 2 addresses change at every network link, while the source and destination IP addresses generally remain unchanged.

10. Explain the routing table codes C, L, S, R, O, and D.

Answer:

Code	Meaning
C	Directly Connected Route
L	Local Route
S	Static Route
R	RIP Route
O	OSPF Route
D	EIGRP Route

Explanation:

- C: A network directly connected to an active router interface.
- L: The specific IP address configured on a router interface.
- S: A route manually configured by an administrator.
- R: A route learned through RIP.
- O: A route learned through OSPF.
- D: A route learned through EIGRP.

11. Explain equal-cost load balancing and convergence.

Answer:**Equal-Cost Load Balancing:**

When multiple routes to the same destination have the same metric, a router may use more than one route to forward traffic. This process is called equal-cost load balancing.

Advantages:

1. Distributes traffic across multiple paths
2. Improves resource utilization
3. Reduces congestion on a single path
4. Can improve network performance

Convergence:

Convergence is the process by which routers exchange updated routing information after a topology change. When all routers have accurate and consistent routing information, the network is said to have converged.